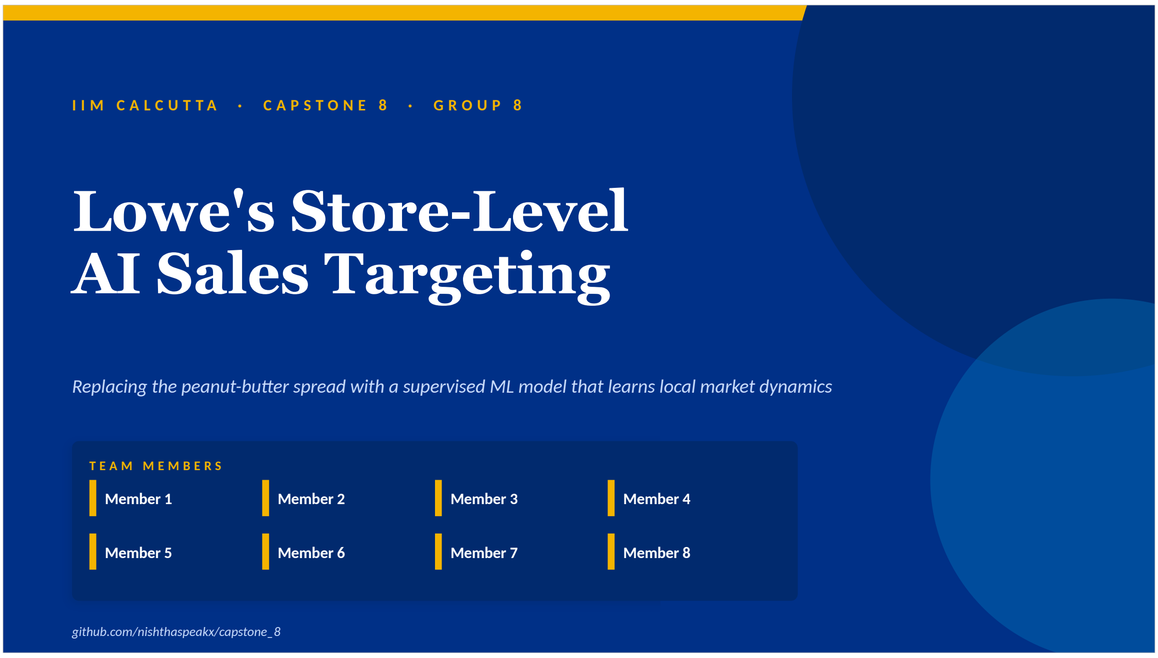


[~15 sec] Good morning Professor. We are Group 8 — and our capstone is on Lowe's Store-Level AI Sales Targeting. Over the next ten minutes, we will walk you through how we replaced their flat 'peanut-butter' planning method with a supervised machine learning model that actually understands local market dynamics. We will spend roughly six minutes on the work, and then move into a live demo of the dashboard for the remaining time.

THE PROBLEM

Today's targets ignore local reality

How Lowe's sets targets today: the total company plan is divided across ~1,727 stores using each store's 3-year historical sales share — a flat allocation that ignores local market reality.

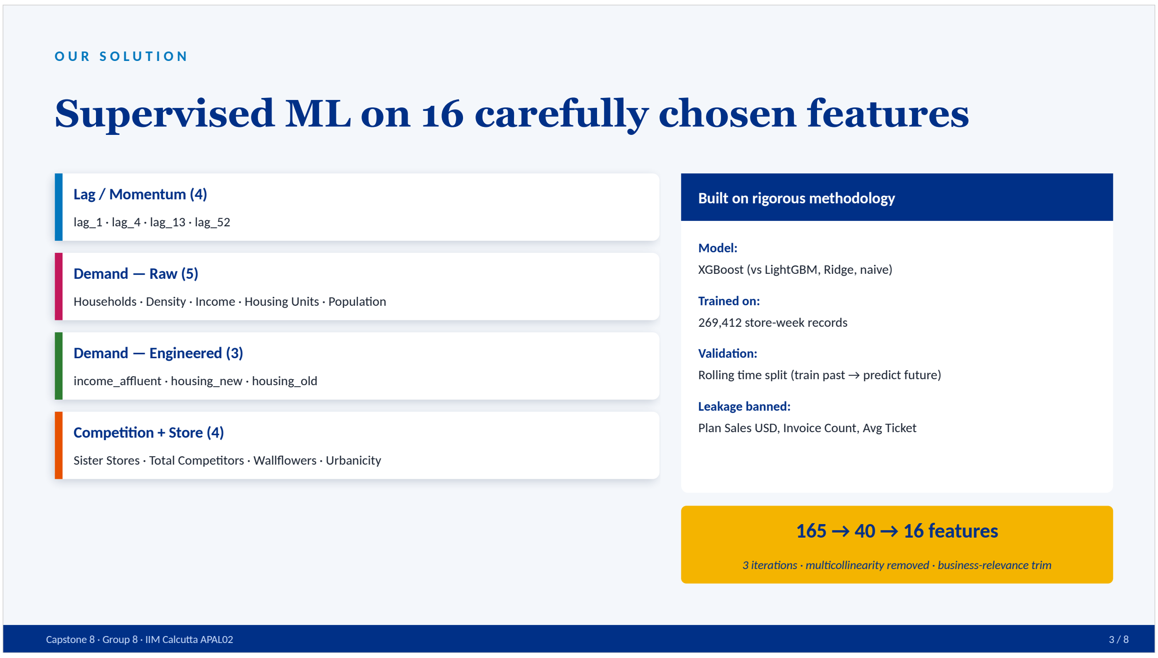
- A booming new-construction suburb gets the SAME growth rate as a declining rural town
- Demographics, housing age, income, competition — all invisible to the method
- Planners spend the whole year manually overriding wrong targets

25.83%

of store-weeks miss target
by more than ±10%
under the current method

1 in 4 store-weeks need re-planning

[~50 sec] Lowe's is America's number two home improvement retailer — about 1,727 stores and 85 billion dollars in annual revenue. Today, when they set next year's store sales targets, they take the total company plan and divide it across stores using each store's 3-year historical sales share. We call this the 'peanut-butter spread' — one flat number smeared across every store. The flaw is obvious: a booming new-construction suburb gets the same growth rate as a declining rural town. Demographics, housing age, income, competition — none of it matters. The cost? Under the current method, 25.83% of store-weeks miss target by more than ten percent. That is one in four store-weeks. Planners spend the entire year doing manual overrides, store by store.



[~55 sec] Our solution is a supervised machine learning model trained on 269,412 store-week records. We started with 165 candidate features, narrowed to 40 in Phase 1, and after three modeling iterations landed on the final 16 you see on the left. They cover four groups: lag features capturing recent momentum, raw demand variables like households and income, three engineered demand features combining things like income brackets and housing age, and competition plus store attributes. The model is XGBoost — gradient boosted trees. We benchmarked it against LightGBM, Ridge regression, and a naive same-week-last-year baseline. Two key methodology points: we used a rolling time split — train on the past, test on the future, never random — and we explicitly banned three columns from being used as inputs because they were same-period measures that would leak the answer.

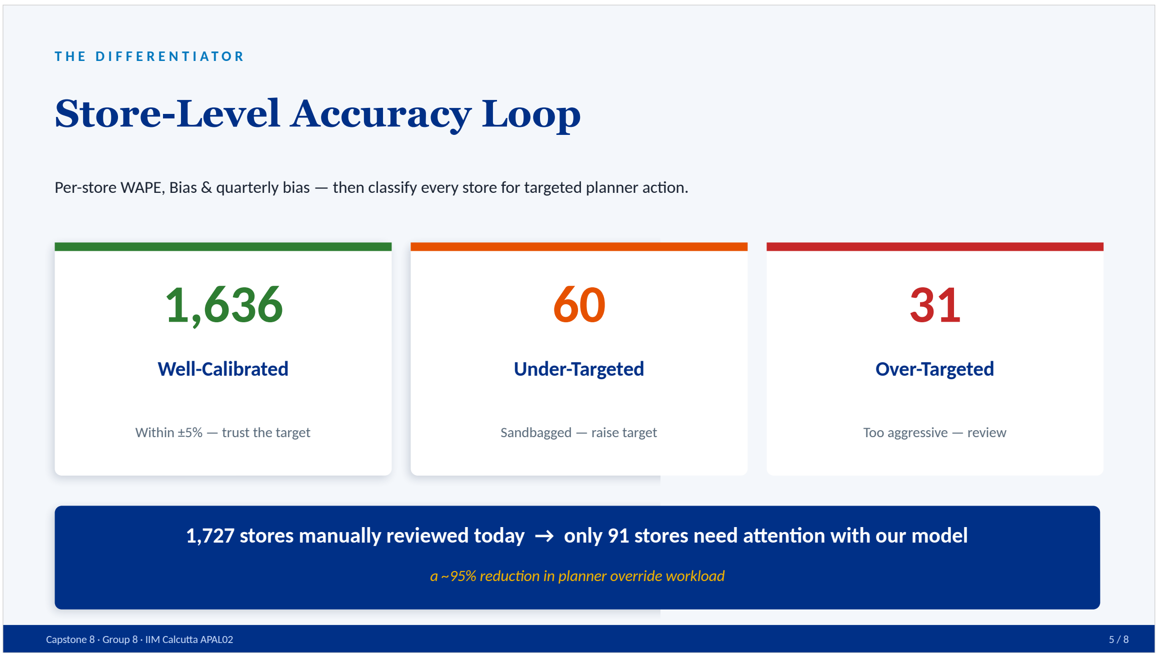
HEADLINE RESULTS

Every success criterion met

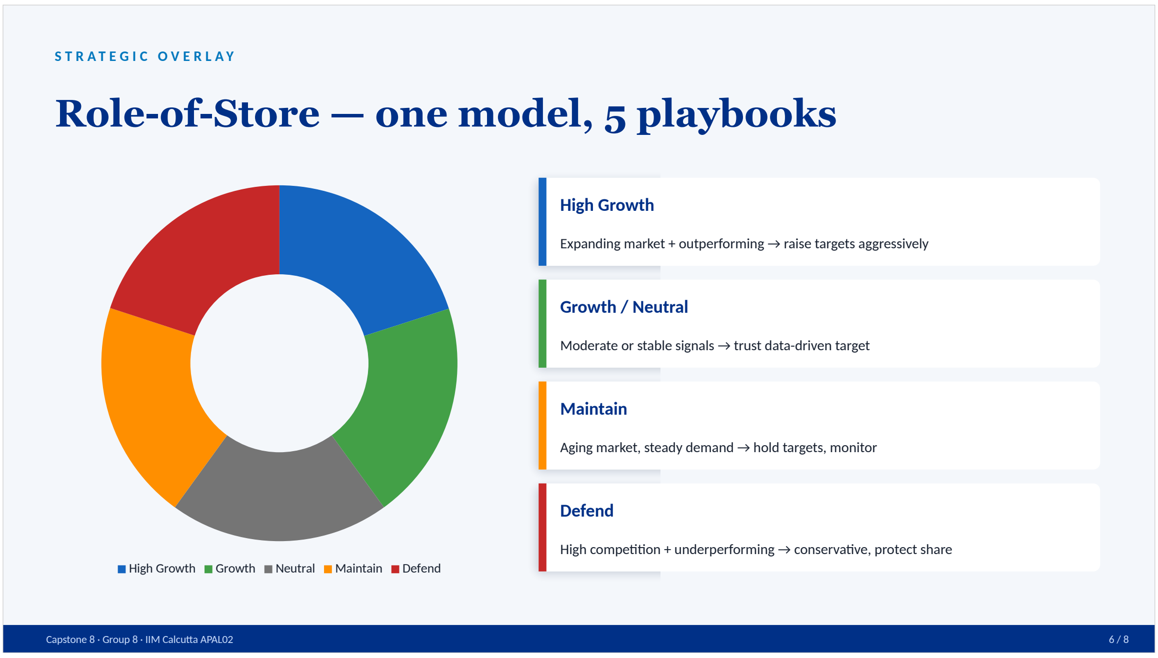
7.05%	−0.97%	0.977	15%	94.7%
WAPE	Bias	R^2	vs Naive	Calibrated
Target < 8%	Within ±2%	Target > 0.88	Beats lag-52	1,636 / 1,727

Model	WAPE	Bias	R^2
XGBoost (selected)	7.05%	−0.97%	0.977
LightGBM	7.07%	−1.11%	0.977
Ridge Regression	7.11%	+0.39%	0.978
Naive (lag-52)	8.27%	+0.33%	0.965

[~50 sec] Every success criterion is met. WAPE of 7.05% — well under our 8% target. WAPE means: of every hundred dollars of actual sales, the model misses by about seven dollars. Bias of minus 0.97% — essentially zero, comfortably within our plus-or-minus 2% acceptance band. That means the chain plan reconciles cleanly. R-squared is 0.977 — the model explains 97.7% of sales variation. We beat the naive baseline by 15% — that is the bar every real model must clear. Look at the comparison table: all three real models cluster around 7% WAPE, while naive is at 8.27%. That agreement tells us the result is robust, not a fluke. The most important number on this slide is 94.7% — the share of stores whose targets land within plus-or-minus 5% of actual. We will unpack that next.



[~55 sec] This is the real differentiator. Instead of trusting one global score that can hide offsetting errors, we compute per-store WAPE and bias, then classify every store. 1,636 stores — 94.7% — are well-calibrated, meaning their target is within plus-or-minus 5% of actual. Planners can trust these directly. Sixty stores are under-targeted: the model predicts too low and the store is essentially sandbagged — planners should raise these targets. Thirty-one are over-targeted: the model is too aggressive and the planner needs to investigate headwinds before committing. The headline business impact is on the navy strip at the bottom: today, planners manually review all 1,727 stores every cycle. With our model, only 91 need human attention. That is a 95% reduction in override workload — and the chain plan still reconciles because bias is near zero.



[~40 sec] On top of the model we layered a Role-of-Store overlay. Every store falls into one of five strategic segments — High Growth, Growth, Neutral, Maintain, or Defend. We derive these from a composite net score that combines household growth, new-housing share, and competitive intensity, then split stores into quintiles. The result: about 20% of stores in each segment, each with its own playbook. High Growth stores get aggressive targets; Defend stores get conservative ones to protect share. This turns a single model into five distinct planning playbooks — giving the planning team a richer toolkit than just one number per store.

REFLECTIONS

What worked, what's honest, what's next

What worked well

- Leakage-safe lag features turned a hard problem tractable
- Trimming to 16 features improved bias AND explainability
- Store-Level Accuracy Loop — the real business differentiator
- 3 model families agreeing at ~7% — robust, not a fluke
- Discipline on leakage (refused Plan Sales USD at corr 0.986)

Honest limitations

- Lag-driven — strong for existing stores, not for new ones
- Week-ahead scoring (no recursive multi-week forecasting yet)
- Role thresholds are heuristic (quantile-based)
- Demographics are annual snapshots; intra-year shifts missed
- No event / promo calendar integrated yet

What's next

- Recursive multi-week forecasting for full-year targets
- Cold-start model for new stores (no lag history)
- Constrained optimization → store sums equal division plan
- Pilot West Division → measure override drop → scale

[~45 sec] Three honest reflections to close. What worked: our discipline on leakage prevention — we refused to use Plan Sales USD even at a 0.986 correlation, because it would have given us fake accuracy but a useless production model. And cutting from 40 features to 16 actually improved bias and made the model far more explainable to planners. Honest limitations: the model is lag-driven, so it is strong for existing stores but weak for brand-new ones. We do not yet have a full-year recursive forecaster, and demographics are annual snapshots that miss intra-year shifts. What is next: a cold-start model for new stores, recursive multi-week forecasting, constrained optimization so store targets sum exactly to the division plan, and a pilot on the West Division to measure override reduction before chain-wide rollout. With that — let me move to the live demo.

Thank You

Questions & discussion welcome

From a flat heuristic → a transparent, store-specific, validated ML targeting system.

7.05%	−0.97%	94.7%	95%
WAPE	Bias	Calibrated	Workload Drop

IIM Calcutta · Capstone 8 · Group 8 | github.com/nishthaspeakx/capstone_8

[~15 sec — show this AFTER the live demo] To recap — 7.05% WAPE, near-zero bias, 94.7% of stores accurately targeted, and a 95% reduction in planner workload. Thank you, Professor. We would love to take your questions.